

ADDITIONAL PROBLEM SHEET 5

Topics covered: Chapter 13.

1. Acetabularia

Consider the following generic model for a reaction–diffusion system contained within an annular domain,

$$\begin{aligned}\frac{\partial u}{\partial t} &= \nabla^2 u + F(u, v), \\ \frac{\partial v}{\partial t} &= D \nabla^2 v + G(u, v).\end{aligned}$$

with $D > 0$ a positive diffusion constant. The domain has a polar coordinate system (r, θ) whose origin is at the focus of the circular boundaries of the domain whose inner and outer radii are R_i and R_o respectively. The system is assumed to have a homogeneous equilibrium (u_0, v_0) . We assume that u and v have no flux boundary conditions at $R = R_i$ and take on their homogeneous equilibrium values at $R = R_o$.

The Laplacian operator in polar coordinates is

$$\nabla^2 u = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2}. \quad (1)$$

- (a) We assume solutions in the form $u(r, \theta, t) = u_0 + \varepsilon u_1(r, \theta, t)$ and $v(r, \theta, t) = u_0 + \varepsilon v_1(r, \theta, t)$. Find the $\mathcal{O}(\varepsilon)$ version of the full equations.
- (b) We assume solutions to this system in the form $u_1 = \psi_u(r, \theta)e^{\lambda t}$ and $v_1 = \psi_v(r, \theta)e^{\lambda t}$, where ψ_u and ψ_v satisfy

$$\nabla^2 \psi_u + k^2 \psi_u = 0, \text{ and } \nabla^2 \psi_v + k^2 \psi_v = 0. \quad (2)$$

Find the general form which ψ_u and ψ_v take.

- (c) Show that if $R_o - R_i$ is $\mathcal{O}(\varepsilon)$ then $k = n/R_i$.

2. Chemotaxis

Consider the following generic model for bacteria in a semi-solid medium:

$$\frac{\partial n}{\partial t} = D_n \nabla^2 n - \alpha \nabla \cdot \left[\frac{n}{(1+c)^2} \nabla c \right] + \rho n \left(\delta \frac{s^2}{1+s^2} - n \right), \quad (3a)$$

$$\frac{\partial c}{\partial t} = D_c \nabla^2 c + \beta s \frac{n^2}{\mu + n^2} - nc, \quad (3b)$$

$$\frac{\partial s}{\partial t} = D_s \nabla^2 s - \kappa n \frac{s^2}{1+s^2}, \quad (3c)$$

where $D_n, D_c, D_s, \alpha, \rho, \delta, \beta, \mu$ and κ are positive constants, n is the bacteria concentration, c is the chemoattractant concentration, and s is the nutrient concentration.

- (a) Describe all the terms in this model.
- (b) State a family of homogeneous equilibria of this system for which $s_0 = \text{const}$ is non-zero.
- (c) Assume s and c are homogeneous and in equilibrium, state why it is not possible to have a non-zero homogeneous equilibrium for n .

3. Chemotaxis and slime mould (Exam section B, 2017)

Consider the following generic model for bacteria in a semi-solid medium:

$$\frac{\partial n}{\partial t} = \nabla^2 n - \alpha \nabla \cdot \left[\frac{n}{(1+c)^2} \nabla c \right] + \rho n \left(\delta \frac{s^2}{1+s^2} - n \right), \quad (4a)$$

$$\frac{\partial c}{\partial t} = D \nabla^2 c + \beta s \frac{n^2}{\mu + n^2} - nc, \quad (4b)$$

$$\frac{\partial s}{\partial t} = D_s \nabla^2 s - \kappa n \frac{s^2}{1+s^2}. \quad (4c)$$

Here $\alpha, \rho, \delta, \beta, \mu, D, D_s$ and κ are positive constants, n is the bacteria concentration, c is the chemoattractant concentration, and s is the nutrient concentration.

- (a) Describe the physical interpretation of all the terms in this model.
- (b) What is the required condition on D for it to represent chemotaxis?
- (c) We assume $\kappa \ll 1$ so that the total nutrition population s remains constant, and we further assume it is homogeneously distributed with a value s_0 . The Turing conditions for this chemotactant system are given by

$$\begin{aligned} F_n + G_c &< 0, \quad F_n G_c - F_c G_n > 0, \\ G_c + D F_n - \alpha \chi_0 G_n &> 0, \quad (G_c + D F_n - \alpha \chi_0 G_n)^2 - 4D(F_n G_c - F_c G_n) \geq 0. \end{aligned}$$

We have used the notation

$$\begin{aligned} F_n &= \left. \frac{\partial F}{\partial n} \right|_{n=n_0, c=c_0}, \quad F_c = \left. \frac{\partial F}{\partial c} \right|_{n=n_0, c=c_0}, \\ G_n &= \left. \frac{\partial G}{\partial n} \right|_{n=n_0, c=c_0}, \quad G_c = \left. \frac{\partial G}{\partial c} \right|_{n=n_0, c=c_0}, \\ \chi_0 &= \chi(n_0, c_0), \end{aligned}$$

where (n_0, c_0) represent a homogeneous equilibrium of the system. Identify the functions F, G and χ in this system.

- (d) Find the permissible homogeneous equilibria (n_0, c_0) for the system (still assuming $s = s_0$ is constant in space and time).
- (e) Show that the first three Turing conditions,

$$\begin{aligned} F_n + G_c &< 0, \quad F_n G_c - F_c G_n > 0, \\ G_c + D F_n - \alpha \chi_0 G_n &> 0, \end{aligned}$$

cannot be satisfied if

$$\delta^2 < \mu. \quad (5)$$

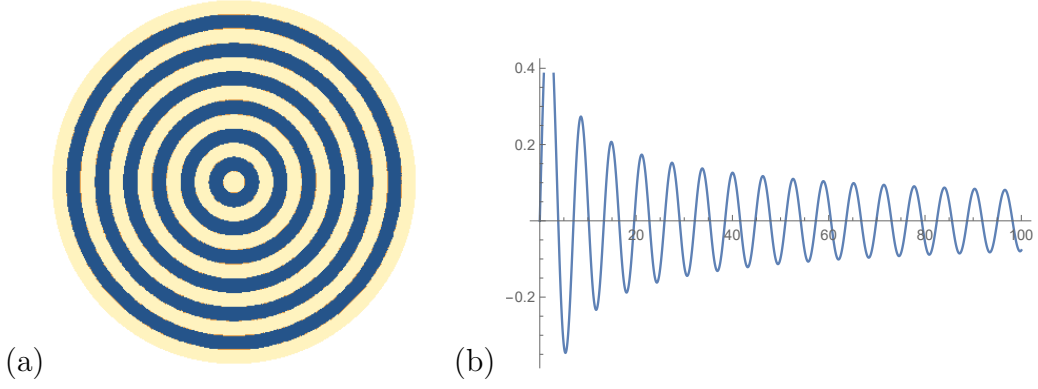


Figure 1: A dartboard like pattern on an annular domain.

4. Non-Euclidean domain

Consider a disk used to model a solid growing bacterial stalk given by a vector $\mathbf{c}(r, \theta)$,

$$\mathbf{c}(r, \theta) = r \cos(\theta) \hat{\mathbf{x}} + r \sin(\theta) \hat{\mathbf{y}}. \quad (6)$$

for $r \in [0, R_0]$ and $\theta \in [0, 2\pi)$, where $(\hat{\mathbf{x}}, \hat{\mathbf{y}})$ are Cartesian basis vectors. In this domain the Laplacian of a function ψ takes the following form

$$\nabla^2 \psi = \frac{\partial^2 \psi}{\partial r^2} + \frac{1}{r} \frac{\partial \psi}{\partial r} + \frac{1}{r^2} \frac{\partial^2 \psi}{\partial \theta^2}. \quad (7)$$

Consider the following generic reaction diffusion type system

$$\begin{aligned} \frac{\partial u}{\partial t} &= \nabla^2 u + F(u, v), \\ \frac{\partial v}{\partial t} &= D \nabla^2 v + G(u, v), \end{aligned}$$

on this domain. There are no-flux boundary conditions on the domain boundary, D is a positive constant.

- State explicitly the boundary conditions for this problem.
- Consider solutions $u(r, \theta, t) = u_0 + \varepsilon u_1(r, \theta, t)$ and $v(r, \theta, t) = v_0 + \varepsilon v_1(r, \theta, t)$, where u_0, v_0 are homogeneous equilibria of the system. Find the $\mathcal{O}(\varepsilon)$ equations and assume their solutions take the form $u_1 = \psi_u(r, \theta) e^{\lambda t}$ and $v_1 = \psi_v(r, \theta) e^{\lambda t}$, where ψ_u and ψ_v satisfy

$$\nabla^2 \psi_u + k^2 \psi_u = 0 \quad \text{and} \quad \nabla^2 \psi_v + k^2 \psi_v = 0. \quad (8)$$

Show that solutions to these eigen-equations take the form

$$\psi_{u/v}(r, \theta) = \sum_{n=0}^{\infty} [f_{n1}(r) \cos(n\theta) + f_{n2}(r) \sin(n\theta)], \quad (9)$$

where $f_{n\alpha}, \alpha = 1, 2$ solve the equation

$$\frac{d^2 f_{n\alpha}}{dr^2} + \frac{1}{r} \frac{df_{n\alpha}}{dr} + \left(k^2 - \frac{n^2}{r^2} \right) f_{n\alpha} = 0. \quad (10)$$

- (c) You are told that the observed patterns form a set of concentric rings, as shown in fig. 1(a). State the form of the permissible spatial patterns ψ_u , given that the solution to eq. (10) takes the form

$$f_{n\alpha}(z) = A_{n\alpha}J_n(kr) + B_{n\alpha}Y_n(kr), \quad (11)$$

with $J_n(x)$ and $Y_n(x)$ the Bessel functions of the first and second kind. The 0th Bessel function of the first kind takes the form

$$J_0(r) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m! \Gamma(m+1)} \left(\frac{r}{2}\right)^{2m}, \quad \Gamma(n) = (n-1)!, \quad (12)$$

and the second kind,

$$Y_0(r) = \frac{2}{\pi} \left\{ [\ln(r/2) + \gamma] J_0(r) + \sum_{k=1}^{\infty} (-1)^{k+1} H_k \frac{(r^2/4)^k}{(k!)^2} \right\}, \quad (13)$$

where γ and H_k are constants. Show that the equation for k simplifies to

$$J_1(kR_0) = 0. \quad (14)$$

- (d) A plot of $J_1(r)$ is shown in fig. 1(b), there is a countable infinity of k values which can solve the boundary condition, eq. (14). Assume there are two stalks whose cross-sectional radii are $R_0 = 1$ and $R_0 = 100$. Assume the reaction–diffusion system is such that the Turing conditions satisfied in both cases and are independent of R_0 . Which domain has more potential patterns?

5. Chemotaxis (Exam section B type)

Consider the following one-dimensional chemotactic slime mould model,

$$\frac{\partial n}{\partial t} = D \frac{\partial^2 n}{\partial x^2} - \alpha \frac{\partial}{\partial x} \left[n \frac{\partial c}{\partial x} \right], \quad (15a)$$

$$\frac{\partial c}{\partial t} = \frac{\partial^2 c}{\partial x^2} + \gamma c \frac{n^2}{\mu + n^2}, \quad (15b)$$

where α, μ are positive constants and $\gamma \in \mathbb{R}$.

- (a) Let $n = n_0$ be a spatially homogeneous equilibrium state for n . Suppose n is in this equilibrium state. If we further assume that c has a spatially homogeneous profile, which initially takes a value c_0 , then show that

$$c = c_0 e^{rt}, \text{ where } r = \gamma \frac{n_0^2}{\mu + n_0^2}. \quad (16)$$

- (b) Consider solutions of the form

$$n(x, t) = n_0 + \varepsilon f(t) \sum_k e^{ikx}, \quad c(x, t) = c_0 e^{rt} + \varepsilon g(t) \sum_k e^{ikx}, \quad (17)$$

where $\varepsilon \ll 1$. Show that, for each k , the $\mathcal{O}(\varepsilon)$ equations are:

$$\frac{\partial f}{\partial t} = -Dk^2 f + \alpha n_0 k^2 g, \quad (18a)$$

$$\frac{\partial g}{\partial t} = -(k^2 - r) g + \frac{2n_0 c_0}{\mu + n_0^2} (\gamma - r) e^{rt} f. \quad (18b)$$

- (c) Show that the system, eq. (18), can be reduced to a single ordinary differential equation for f in the form

$$\frac{d^2 f}{dt^2} + a(k) \frac{df}{dt} + [b(k)e^{rt} + c(k)] f = 0, \quad (19)$$

where

$$a(k) = [Dk^2 + (k^2 - r)], \quad b(k) = \alpha n_0 \beta (r - \gamma) k^2, \quad \beta = \frac{2n_0 c_0}{\mu + n_0^2}, \quad (20)$$

$$c(k) = (k^2 - r) Dk^2. \quad (21)$$

- (d) One technique for assessing the behaviour of each mode could be to make the assumption that the coefficients of the equation

$$\frac{d^2 f}{dt^2} + a(k) \frac{df}{dt} + [b(k)e^{rt} + c(k)] f = 0 \quad (22)$$

vary at a rate slower than the function f and its derivatives. Why is this assumption incorrect in this case?

- (e) We now assume $r/\gamma = 1$ and consider the problem on a domain of length L with no-flux boundary conditions. State a condition for which all growth modes will be stable.

6. Dynamic Turing analysis

Consider the following model for the interaction of a bacterial species $u(x, t)$ and a chemical density $v(x, t)$:

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} - \frac{\partial}{\partial x} \left[\chi(u, v) \frac{\partial v}{\partial x} \right] + u(1 - u), \quad (23a)$$

$$\frac{\partial v}{\partial t} = D \frac{\partial^2 v}{\partial x^2} + uv \left(\frac{1}{t} - 1 \right), \quad (23b)$$

for $t \in (0, \infty]$, $x \in [0, L]$ and both subject to no-flux boundary conditions. Here D is a constant and $\chi(u, v)$ a smooth scalar function.

- (a) Describe the physical interpretation of the various terms on the RHS of eq. (23).
 (b) Assume u is in a stable homogeneous equilibrium. Assume further that v remains homogeneous. Solve for the temporal growth of $v(t)$. You may assume any results derived in your class notes.
 (c) We now seek to assess whether temporally transient patterns can form in the system. Assume solutions to eq. (23) in the form

$$u(x, t) = 1 + \varepsilon u_1(x, t) = 1 + \varepsilon \sum_{n=0}^{\infty} f_n(t) e^{ik_n x}, \quad (24a)$$

$$v(x, t) = v_0(t) + \varepsilon v_1(x, t) = v_0(t) + \varepsilon \sum_{n=0}^{\infty} g_n(t) e^{ik_n x}, \quad (24b)$$

where, $k_n = n\pi/L$, $\varepsilon \ll 1$, and $v_0(t)$ is the homogeneous growth function found in part (b). Show the order ε equations take the form:

$$\begin{aligned}\frac{df_n}{dt} &= -k_n^2 f_n + k_n^2 \chi(u_0, v_0) g_n - f_n, \\ \frac{dg_n}{dt} &= -Dk_n^2 g_n + (g_n + f_n v_0(t)) \left(\frac{1}{t} - 1 \right).\end{aligned}$$

- (d) Assume $\chi(u, v) = 1$. Demonstrate that the system *can* exhibit transient patten formation under the assumption of solutions in the form of eq. (24).

Hint: You do not need to solve for f_n and g_n for general t , just consider their approximate behaviour at small t and large t . You may assume the solutions to g_n and f_n remain > 0 for all t .